References

1. Rizzo JF III, Andreoli CM, Rabinov JD. Use of magnetic resonance imaging to differentiate optic neuritis and nonarteritic anterior ischemic optic neuropathy. Ophthalmology 2002;109:1679–84.
2. Lee AG, Eggenberger ER, Kaufman DI, Manrique C. Optic nerve enhancement on magnetic resonance imaging in arteritic ischemic optic neuropathy. J Neuroophthalmology 1999;19: 235–7.

Author reply

Dear Editor:
With regard to Dr. Lee's question of a possible alternative diagnosis of arteritis in patients with enhancing magnetic resonance imaging lesions whom we presumed to have nonarteritic anterior ischemic optic neuropathy (NAION), we were mindful to exclude cases that were even suggestive of arteritis. Our description of methods was intended to make clear our attempt to identify and eliminate such cases.

We also were curious about the two cases of "NAION" that had enhancement of the optic nerve, which prompted us to review these cases again. Of the two patients with enhancement, neither reported symptoms consistent with giant cell arteritis or other causes of inflammatory or infectious optic neuropathy. The erythrocyte sedimentation rates were 7 and 22, respectively, and complete blood counts were normal. Angiotensin-converting enzyme, antinuclear antibody, and fluorescent treponemal antibody blood test results were normal in one patient (none of these were obtained in the other). Both patients had small cup/disc ratios. Follow-up examination did not uncover evidence of diagnoses more likely than NAION.

Interestingly, the magnetic resonance imaging scan of patient 56 was interpreted as normal by both the senior author and a neuroradiologist (other than the study neuroradiologist). Our study neuroradiologist (JDR) interpreted the scan as showing a small area of "slight enhancement" of the clinically affected optic nerve, which according to the design of the study was recorded as a positive scan. By coincidence, this scan was randomly selected for a re-review (by the same neuroradiologist) to determine the degree of consistency of interpretations, and the second interpretation was the same (i.e., slight enhancement). The variance among reviewers indicates that this abnormality was subtle and perhaps of questionable clinical significance.

We do not have an explanation for the apparent enhancement of the optic nerve on magnetic resonance imaging scans of patients who are presumed to have NAION. We accept the possibility that the radiologic findings might indicate an alternative diagnosis, although none has become evident in these two patients.

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Retinal Hemorrhages in Children

Dear Editor:
In Schloff et al's article,[1] all cases of suspected nonaccidental trauma were excluded. Because two of the authors in a previous article stated the finding of retinal hemorrhages in a child with a head injury suggests a nonaccidental cause,[2] the conclusion that retinal hemorrhage is uncommon in children with intracranial hemorrhage not resulting from shaken baby syndrome is unwarranted. The only trauma cases left in the study should have been those without retinal hemorrhage. Indeed, the 7-year-old should probably have been excluded, although the earlier article only studied children younger than 3. The correct conclusion is that very few children with retinal hemorrhage cannot be suspected as having been abused.

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References

1. Schloff S, Mullaney PB, Armstrong DC, et al. Retinal findings in children with intracranial hemorrhage. Ophthalmology 2002; 109:1472–6.
2. Buys YM, Levin AV, Enzenauer RW, et al. Retinal findings after head trauma in infants and young children. Ophthalmology 1992;99:1718–23.

Author reply

Dear Editor:
Although we welcome Dr. Gardner's letter, we found it difficult to interpret his comments. We think he is concerned that the children in our study were preselected in such a way as to eliminate those children with retinal hemorrhages who might have been assumed to have been victims of the shaken baby syndrome.

We did exclude all cases of suspected nonaccidental trauma. However, this was done by the admitting hospital service *before* the child was examined by the ophthalmology team. Therefore, retinal hemorrhages were not used to predetermine the diagnosis. Interestingly, those children who were excluded from our study based on an initial suspicion of possible abuse (all children had their eye examinations within 72 hours of admission) but whose cases were later adjudicated by our child abuse team as accidental injuries had no retinal hemorrhages. Had they been included in our study, this would have further underscored the rarity of retinal hemorrhage associated with intracranial hemorrhage in children who are not victims of shaken baby syndrome.

Dr. Gardner refers to another publication from our center.[1] Although the intent of that study was not to specifically address the issue of retinal findings in children with intracranial hemorrhage, of the 79 children with head injury, 8 had intracranial hemorrhage. Three of those eight cases were nonaccidental trauma, and all had retinal hemorrhages. The remaining five cases of accidental trauma all showed no evidence of retinal hemorrhages, again supporting our conclusion that "retinal hemorrhage is uncommon in children with intracranial hemorrhage" not due to child abuse.

We agree that the diagnosis of shaken baby syndrome